STALKS OF EXTENSION BY ZERO ALONG AN ETALE MORPHISM, AND ITS EXACTNESS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both the displayed formula and the exactness of the functor.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Extension by zero j_! along an etale morphism. Same chapter, the functor j_! on abelian sheaves and its description by sections with support in the image of j. Supplies the functor whose stalks are computed.
- Stalk at a geometric point and the description of sections of extension by zero. Same chapter, the description of the sections of j_!F as sums of sections of F supported in the image, and the definition of the stalk as a filtered colimit. Supplies the computation in the proof and the exactness conclusion.

Source reference. Original lemma 03DJ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11439–11439. Navigation pointer invoked at proof line 11439; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0710: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11442–11442. Navigation pointer invoked at proof line 11442; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 04HU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3610–3610. Navigation pointer invoked at proof line 11444; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Proposition 0.1. *Let $j : U \to X$ be an étale morphism of schemes. Let $\mathcal{F}$ in $Ab(U_{étale})$. If $\overline{x} : \mathrm{Spec}(k) \to X$ is a geometric point of X, then*

$$(j_!\mathcal{F})_{\overline{x}} = \bigoplus\nolimits_{\overline{u}:\mathrm{Spec}(k)\to U,\ j(\overline{u})=\overline{x}} \mathcal{F}_{\overline{u}}.$$

In particular, $j_!$ is an exact functor.

Proof. Exactness of $j_!$ is very general, see Modules on Sites, Lemma 03DJ. Of course it does also follow from the description of stalks. The formula for the stalk follows from Modules on Sites, Lemma 0710 and the description of points of the small étale site in terms of geometric points, see Lemma 04HU.

For later use we note that the isomorphism

$$\begin{aligned}(j_!\mathcal{F})_{\overline{x}} &= (j_{p!}\mathcal{F})_{\overline{x}} \\ &= \mathrm{colim}_{(V,\overline{v})}\, j_{p!}\mathcal{F}(V) \\ &= \mathrm{colim}_{(V,\overline{v})} \bigoplus\nolimits_{\varphi:V\to U} \mathcal{F}(V \xrightarrow{\varphi} U) \\ &\to \bigoplus\nolimits_{\overline{u}:\mathrm{Spec}(k)\to U,\ j(\overline{u})=\overline{x}} \mathcal{F}_{\overline{u}}.\end{aligned}$$

constructed in Modules on Sites, Lemma 0710 sends $(V, \overline{v}, \varphi, s)$ to the class of s in the stalk of $\mathcal{F}$ at $\overline{u} = \varphi(\overline{v})$. □